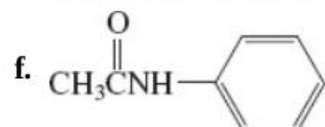
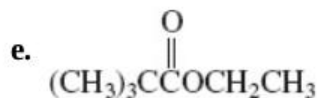
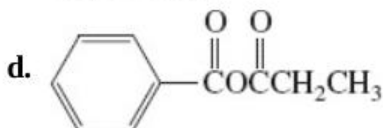
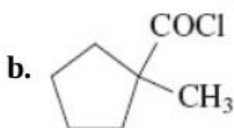
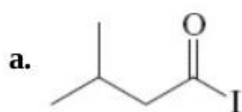


Problems

30. Name (IUPAC system) or draw the structure of each of the following molecules.



g. Propyl butanoate

h. Butyl propanoate

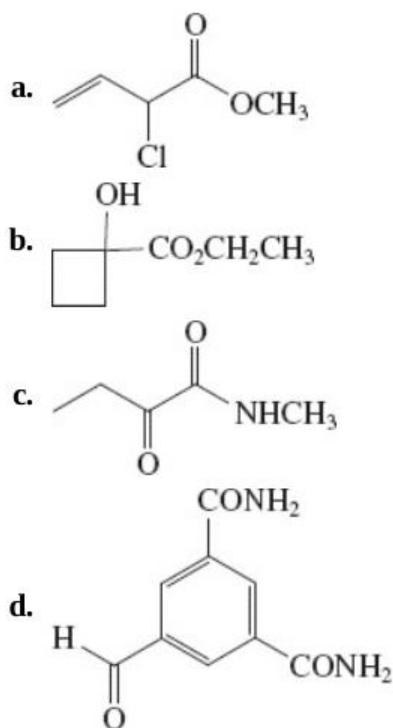
i. 2-Chloroethyl benzoate

j. *N,N*-Dimethylbenzamide

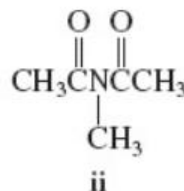
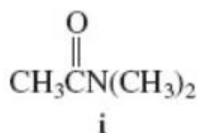
k. 2-Methylhexanenitrile

l. Cyclopentanecarbonitrile

31. Name each of the following structures according to IUPAC or *Chemical Abstracts* rules. Pay attention to the order of precedence of the functional groups.



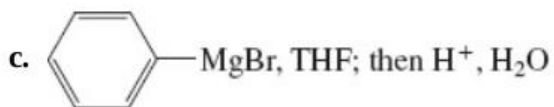
32. **(a)** Use resonance forms to explain in detail the relative order of acidity of carboxylic acid derivatives, as presented in [Section 20-1](#). **(b)** Do the same, but use an argument based on inductive effects.
33. In each of the following pairs of compounds, decide which possesses the indicated property to the greater degree. **(a)** Length of C–X–X bond: acetyl fluoride or acetyl chloride. **(b)** Acidity of the boldface H: $\text{CH}_2(\text{COCH}_3)_2$ **CH₂**(COCH₃)₂ or $\text{CH}_2(\text{COOCH}_3)_2$ **CH₂**(COOCH₃)₂. **(c)** Reactivity toward addition of a nucleophile: (i) an amide or (ii) an imide (as shown below). **(d)** High-energy infrared carbonyl stretching frequency: ethyl acetate or ethenyl acetate.



34. Give the product(s) of each of the following reactions.



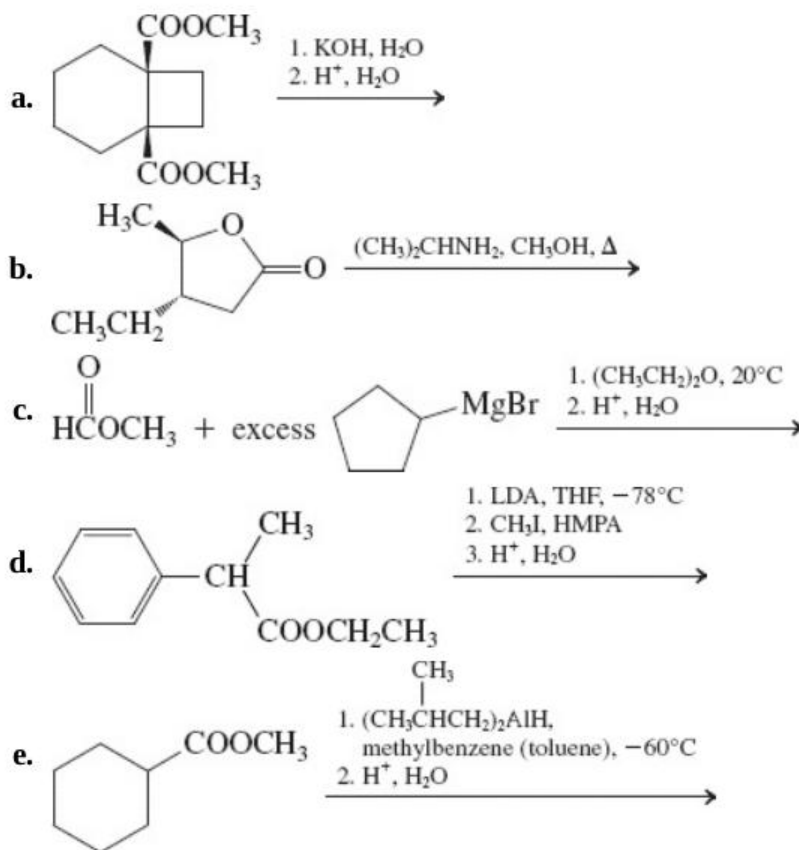
- a. $(\text{CH}_3)_2\text{CHOH}(\text{CH}_3)_2\text{CHOH}$
 b. NH_3NH_3
 c.  $\text{MgBr, THF; then H}^+, \text{H}_2\text{O}$



- d. $\text{LiAlH}_4, (\text{CH}_3\text{CH}_2)_2\text{O}; \text{LiAlH}_4, (\text{CH}_3\text{CH}_2)_2\text{O};$ then $\text{H}^+, \text{H}_2\text{O}$
37. Give the product(s) of the reaction of butanedioic (succinic) anhydride with each of the reagents in [Problem 36](#).
38. Formulate a mechanism for the reaction of butanedioic (succinic) anhydride with methanol shown on [p. 978](#).
39. Give the products of reaction of methyl pentanoate with each of the following reagents under the conditions shown.
- a. $\text{NaOH}, \text{H}_2\text{O}, \text{H}_2\text{O}, \text{heat};$ then $\text{H}^+, \text{H}_2\text{O}$
- b. $(\text{CH}_3)_2\text{CHCH}_2\text{CH}_2\text{OH}, (\text{CH}_3)_2\text{CHCH}_2\text{CH}_2\text{OH}$ (excess), H^+
- c. $(\text{CH}_3\text{CH}_2)_2\text{NH}, (\text{CH}_3\text{CH}_2)_2\text{NH}, \text{heat}$
- d. $\text{CH}_3\text{MgI}, \text{CH}_3\text{MgI}$ (excess), $(\text{CH}_3\text{CH}_2)_2\text{O}; (\text{CH}_3\text{CH}_2)_2\text{O};$ then $\text{H}^+, \text{H}_2\text{O}$
- e. $\text{LiAlH}_4, (\text{CH}_3\text{CH}_2)_2\text{O}; \text{LiAlH}_4, (\text{CH}_3\text{CH}_2)_2\text{O};$ then $\text{H}^+, \text{H}_2\text{O}$
- f. $[(\text{CH}_3)_2\text{CHCH}_2]_2\text{AlH}, [(\text{CH}_3)_2\text{CHCH}_2]_2\text{AlH},$ methylbenzene (toluene), low temperature; then $\text{H}^+, \text{H}_2\text{O}$
40. Give the products of reaction of γ -valerolactone (5-methyloxa-2-cyclopentanone, [Section 20-4](#)) with each of the reagents in [Problem 39](#).
41. Draw the structure of each of the following compounds: (a) β -butyrolactone; (b) β -valerolactone; (c) δ -valerolactone; (d) β -propiolactam; (e) α -methyl- δ -valerolactam; (f) *N*-methyl- γ -butyrolactam.
42. Formulate a mechanism for the acid-catalyzed transesterification of ethyl 2-methylpropanoate (ethyl isobutyrate) into the corresponding methyl ester. Your mechanism should clearly illustrate the catalytic role of the proton.
43. Formulate a mechanism for the reaction of methyl 9-octadecenoate with 1-dodecanamine shown on [p. 984](#).

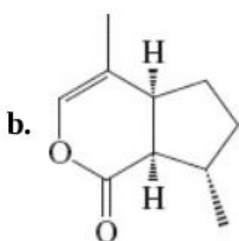
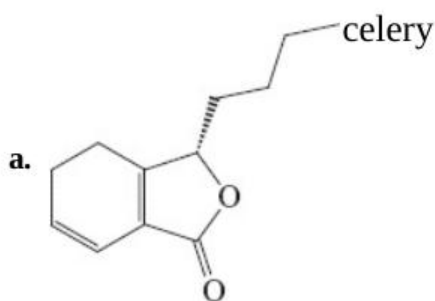
44. Reaction review. Suggest reagents to convert each of the following starting materials into the indicated product: **(a)** acetyl chloride into acetic hexanoic anhydride; **(b)** methyl hexanoate into *N*-methylhexanamide; **(c)** hexanoyl chloride into hexanal; **(d)** hexanenitrile into hexanoic acid; **(e)** hexanamide into hexanamine; **(f)** hexanamide into pentanamine; **(g)** ethyl hexanoate into 3-ethyl-3-octanol; **(h)** hexanenitrile into 1-phenyl-1-hexanone [$\text{C}_6\text{H}_5\text{CO}(\text{CH}_2)_4\text{CH}_3$].

45. Give the product of each of the following reactions.

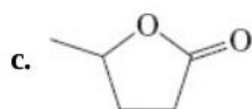


46. For each of the naturally occurring lactones below, draw the structure of the compound that would result from hydrolysis using aqueous base.

Sedanenolide, major contributor to the flavor of



Nepetalactone [[Problem 54, part \(b\) of Chapter 19](#)], major active component in catnip (**Caution:** Do you notice anything unusual about one of the functional groups after hydrolysis?)



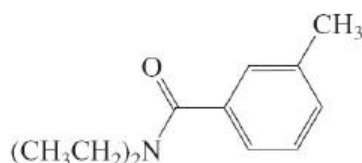
γ -Valerolactone, important in the fragrance industry and a potential biofuel

- d. A novel “green” process converts levulinic acid (4-oxopentanoic acid, $\text{CH}_3\text{COCH}_2\text{CH}_2\text{COOH}$), which is readily obtained from waste biomass, into γ -valerolactone by treatment with gaseous H_2 in the presence of a special hydrogenation catalyst. Speculate on how this reaction may take place.

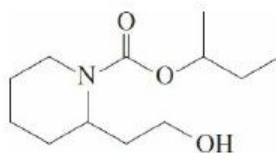
47.

a. *N,N*-Diethyl-3-methylbenzamide (*N,N*-diethyl-*m*-toluamide), marketed as DEET, is perhaps the most widely used insect repellent in the world and is especially effective in interrupting the transmission of diseases by mosquitoes and ticks. Propose one or more preparations of DEET (see structure below) from 3-methylbenzoic acid and any other appropriate reagents.

b. Icaridin (also called picaridin, see structure below) is another repellent useful against biting bugs. Like DEET, icaridin disrupts mosquitoes' odorant receptors, rendering them incapable of detecting humans. What is the name of the carbonyl-containing functional group in icaridin? What would you expect of the reactivity of its carbonyl group compared to that of the other carboxylic acid derivatives discussed in this chapter? Explain your answer.



DEET



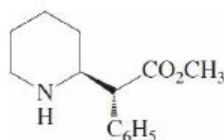
Icaridin

(commercial product
is a mixture of all
four stereoisomers)

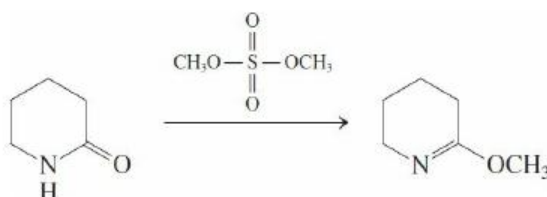
c. Predict the products of hydrolysis of icaridin under basic conditions. (**Hint:** Examine the process mechanistically and look for similarities to a species introduced in [Section 20-7](#).)

48.

a. Like many chiral pharmaceuticals, Ritalin, widely used to treat attention deficit disorder, is synthesized and marketed as a racemic mixture. A synthetic route to the active enantiomer (see structure below) begins with the reaction of dimethyl sulfate with a six-membered ring lactam. Propose a mechanism for this reaction.



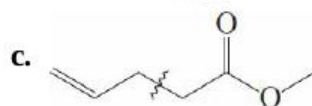
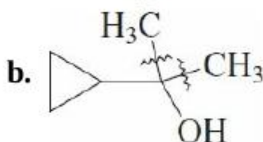
Ritalin
(Active stereoisomer)



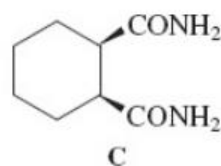
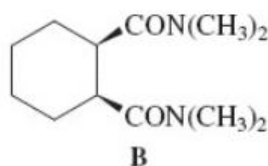
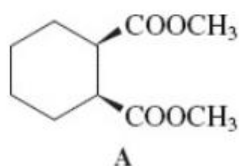
b. How many stereocenters does Ritalin contain? What are their stereochemical designations (*R* or *S*)? How many stereoisomers does Ritalin have, and how are they structurally related?

49. Formulate a mechanism for the formation of acetamide, $\text{CH}_3\text{C}(=\text{O})\text{NH}_2$, from methyl acetate and ammonia.

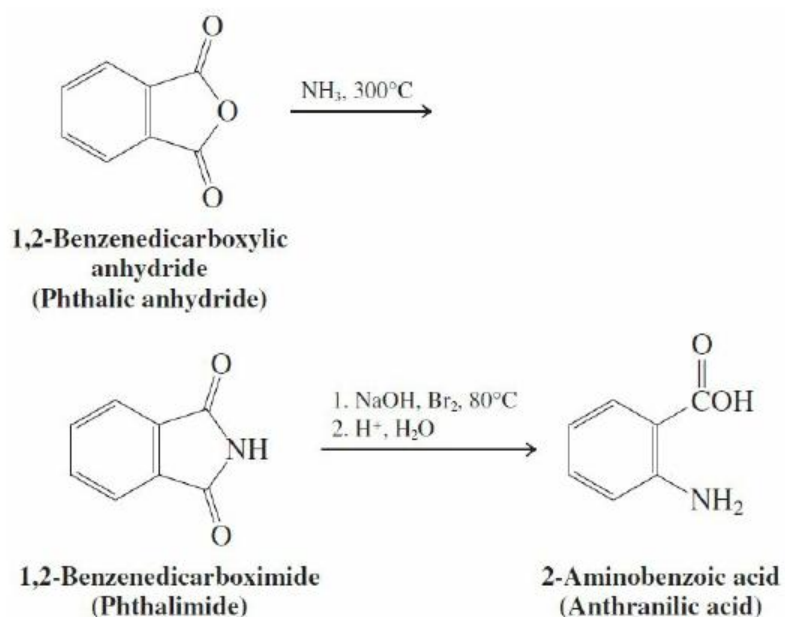
50. Give the products of the reactions of pentanamide with the reagents given in [Problem 39](#), parts (a, e, f). Repeat for *N,N*-dimethylpentanamide.
51. Formulate a mechanism for the acid-catalyzed hydrolysis of 3-methylpentanamide shown on [p. 991](#). (**Hint:** Use as a model the mechanism for general acid-catalyzed addition–elimination presented in [Section 19-7](#).)
52. What reagents would be necessary to carry out the following transformations? **(a)** Cyclohexanecarbonyl chloride $\rightarrow$ 1-cyclohexyl-1-pentanone; **(b)** 2-butenedioic (maleic) anhydride $\rightarrow$ (Z)-2-butene-1,4-diol; **(c)** 3-methylbutanoyl bromide $\rightarrow$ 3-methylbutanal; **(d)** benzamide $\rightarrow$ 1-phenylmethanamine; **(e)** propanenitrile $\rightarrow$ 3-hexanone; **(f)** methyl propanoate $\rightarrow$ 4-ethyl-4-heptanol.
53. In each of the following retrosynthetic disconnections ($\sim$), show reactions that would make the indicated C–C–C bonds.



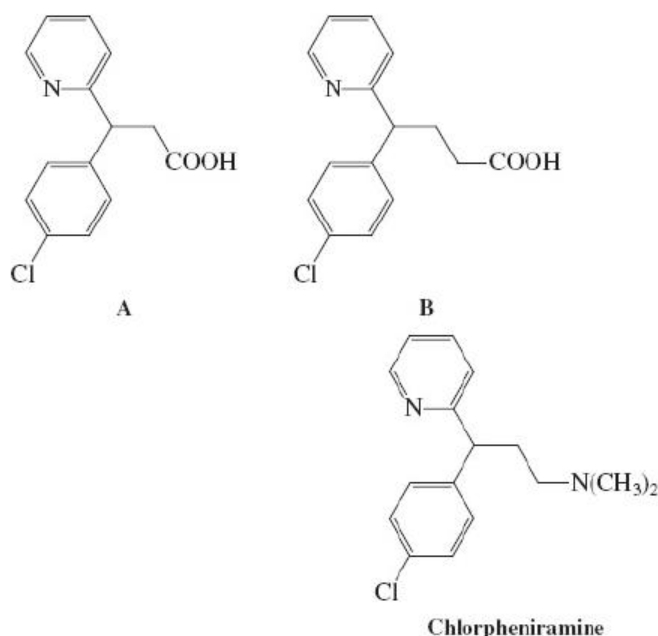
54. Upon treatment with LDA followed by protonation, compounds A and B undergo cis-trans isomerization, but compound C does not. Explain.



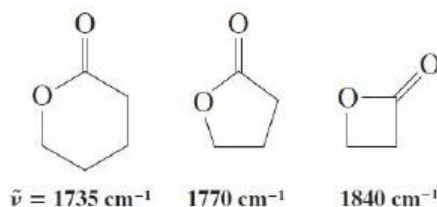
55. 2-Aminobenzoic (anthranilic) acid is prepared from 1,2-benzenedicarboxylic anhydride (phthalic anhydride) by using the two reactions shown here. Explain these processes mechanistically.



56. On the basis of the reactions presented in this chapter, write reaction summary charts for esters and amides similar to the chart for acyl halides ([Figure 20-1](#)). Compare the number of reactions for each of the compound classes. Is this information consistent with your understanding of the relative reactivity of each of the functional groups?
57. Show how you might synthesize chlorpheniramine (see also [p. 762](#)), a powerful antihistamine used in several decongestants, from each of carboxylic acids A and B. Use a different carboxylic amide in each synthesis.



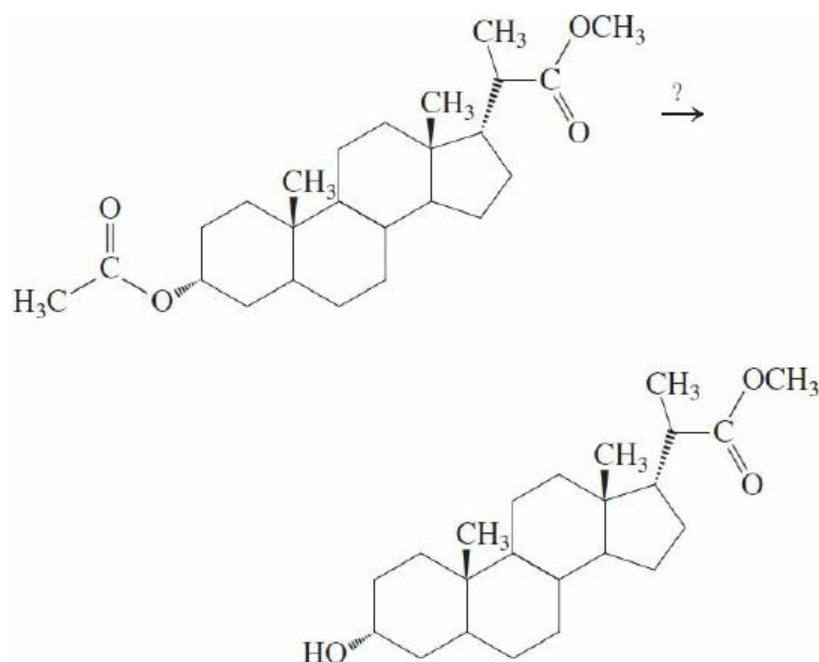
58. Although esters typically have carbonyl stretching frequencies at about $\tilde{\nu} = 1740 \text{ cm}^{-1}$ in the infrared spectrum, the corresponding value for lactones can vary greatly with ring size. Three examples are shown below. Propose an explanation for the increasing energies of the carbonyl stretching vibrations with decreasing ring size of the lactone.



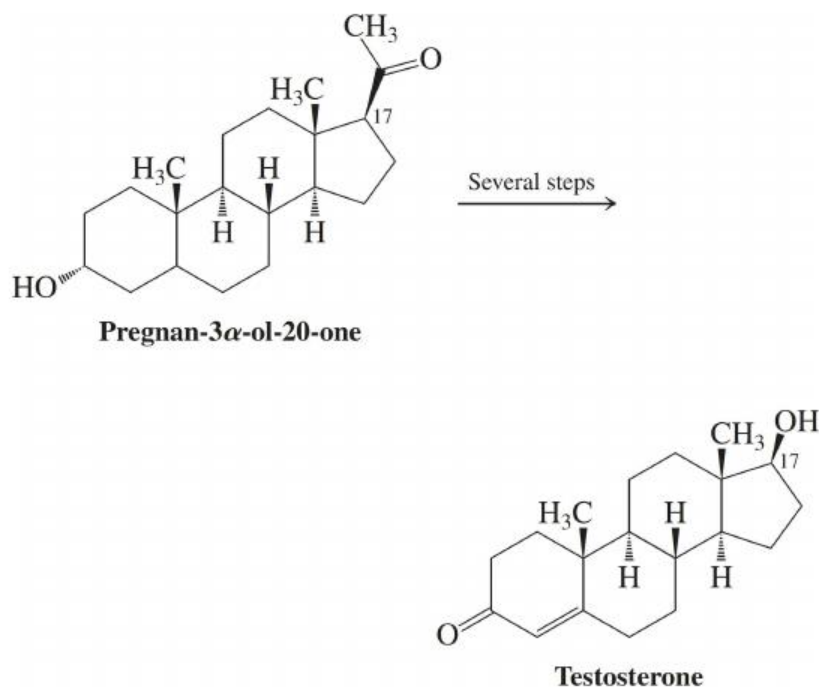
59. Upon completing a synthetic procedure, every chemist is faced with the job of cleaning glassware. Because the materials present may be dangerous in some way or have unpleasant properties, a little serious chemical thinking is often beneficial before “doing the dishes.” Suppose that you have just completed a synthesis of hexanoyl chloride, perhaps to carry out the reaction in [Problem 34](#), part (b); first, however, you must clean the glassware contaminated with this acyl halide. Both hexanoyl chloride and hexanoic acid have terrible odors. **(a)** Would initial cleansing of the glassware with soap and water be a good idea?

Explain. **(b)** Suggest a more agreeable alternative, based on the chemistry of acyl halides and the physical properties (particularly the odors) of the various carboxylic acid derivatives.

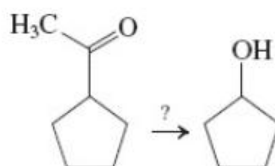
60. **CHALLENGE** Show how you would carry out the following transformation in which the ester function at the lower left of the molecule is converted into a hydroxy group but that at the upper right is preserved. (**Hint:** Do not try ester hydrolysis. Look carefully at how the ester groups are linked to the steroid and consider an approach based on transesterification.)



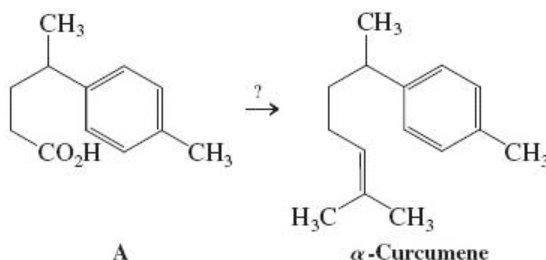
61. The removal of the C17 side chain of certain steroids is a critical element in the synthesis of a number of hormones, such as testosterone, from steroids in the relatively readily available pregnane family.



How would you carry out the comparable transformation, shown next, of 1-cyclopentyl-1-ethanone into cyclopentanol? (*Note:* In this and subsequent synthetic problems, you may need to use reactions from several areas of carbonyl chemistry discussed in [Chapters 17-20](#).)



62. Propose a synthetic sequence to convert carboxylic acid A into the naturally occurring sesquiterpene $\alpha\alpha$ -curcumene.



63. Propose a synthetic scheme for the conversion of lactone A into amine